Evidence Base for Explainable Music Features

Stable feature contract, acoustic interpretation, and research justification

CURRENT STABLE VERSION

41 instrument features • 10 rhythm features • 12 timbre features • 12 harmony features • 6 genre targets

Purpose. This document records why each interpretable timbre and harmony descriptor is relevant to music-genre analysis, ties the feature families to published music-information-retrieval (MIR) research, and fixes the boundary between warehouse candidates and the stable model input.

Evidence claim. The cited literature supports the acoustic construct and its use in music analysis; it does not prove that this exact 12-feature subset is universally optimal. The final subset is a transparent, low-dimensional engineering choice balancing coverage, interpretability, redundancy, and reproducibility.

1. Stable feature contract

Branch	Count	Interpretive scope	Status
Instrument	41	Recognizable source/instrument concepts	Stable input
Rhythm	10	Tempo, beat regularity, onset activity, danceability	Stable input
Timbre	12	Brightness, spread, shape, contrast, noisiness, change, harmonicity	Stable input
Harmony	12	Key/mode, tonal stability, chroma organization, movement, recurrence, chords	Stable input
Genre	6	Multilabel genre targets	Stable output
Harmony warehouse	15	Stable 12 plus three retained variability candidates	Warehouse only

Strict boundary: the 15 harmony fields are retained only in the feature warehouse. The current stable harmony branch uses exactly 12 features; the three additional warehouse fields are not part of the stable vector.

2. Selection logic

- Construct coverage: retain features that expose different audible dimensions instead of many near-duplicates.
- Human-readable direction: a higher or lower value should permit a defensible acoustic interpretation.
- Temporal information: pair selected central-tendency measures with variability or movement measures where variation is musically meaningful.
- Published provenance: use established MIR descriptors or transparent statistics derived from established representations.
- Stable contract: warehouse availability does not automatically authorize use as a model input.

3. Timbre branch — stable 12

Timbre is multidimensional: it reflects spectral distribution, temporal evolution, and harmonic versus noise-like structure. The selected descriptors deliberately avoid an opaque all-MFCC representation. Peeters et al. organize timbre descriptors across temporal, spectral, spectro-temporal, energetic, and harmonic dimensions [1]; Tzanetakis and Cook show the value of short-time spectral statistics for genre classification [2].

Feature	What it measures	Readable meaning	Evidence and study fit
spectral_centroid_mean	Average spectral centre of gravity.	Overall brightness: more high-frequency energy generally raises the value.	Established perceptual correlate of brightness and a standard timbral-texture descriptor [1–3].
spectral_centroid_std	Across-frame variability of centroid.	How much brightness changes through the track.	Temporal summary statistics preserve timbral evolution that a single mean loses [1,2].
spectral_bandwidth_mean	Average spread around the centroid.	Broad/full versus narrow/concentrated spectral energy.	Spectral spread is a core spectral-shape dimension in systematic timbre descriptor sets [1].
spectral_bandwidth_std	Across-frame variability of bandwidth.	Changes in spectral fullness or orchestration density.	Complements mean spread with temporal variability, consistent with frame-distribution summaries [1,2].
spectral_skewness_mean	Average asymmetry of the magnitude spectrum.	Whether the spectral tail is biased toward lower or higher frequencies.	Higher-order spectral shape is included in the Timbre Toolbox taxonomy and adds shape information beyond centre and spread [1].
spectral_contrast_mean	Average peak–valley contrast across octave bands.	Separation of prominent harmonic peaks from spectral valleys/noise floor.	Proposed specifically for music-type classification; shown to retain relative spectral structure lost by average envelopes [4].
spectral_flatness_mean	Geometric-to-arithmetic spectral energy ratio.	Noise-like/flat versus tonal/peaked spectrum.	A standard descriptor for distinguishing noise-like and harmonic spectra in timbre analysis [1].
spectral_rolloff_mean	Frequency below which a fixed proportion of spectral energy lies.	High-frequency extent and spectral edge.	A canonical timbral-texture feature used in genre classification [2] and included in broad timbre descriptor taxonomies [1].
spectral_flux_mean	Average frame-to-frame spectral change.	Typical rate/amount of timbral activity and note/onset-related change.	Used as a timbral-texture statistic in genre classification and linked to perceived activity in polyphonic timbre [2,5].
spectral_flux_std	Variability of frame-to-frame spectral change.	Whether timbral activity is steady or alternates between stable and changing passages.	Adds dynamic dispersion to the established spectral-flux construct [2,5].
hnr_mean_db	Mean harmonic-to-noise ratio.	Periodic/tonal dominance versus noisy or breathy energy.	Harmonic/noise balance is an interpretable component of timbre and is represented in comprehensive timbre descriptor families [1].
inharmonicity_mean	Mean deviation of partials from integer multiples of a fundamental.	Harmonic purity versus stretched, metallic, bell-like, or percussive spectra.	Inharmonicity is an established harmonic spectral descriptor in the Timbre Toolbox [1].

Coverage judgement

Together, the Timbre-12 cover seven complementary axes: brightness (centroid), spread (bandwidth), shape (skewness), local peak/valley structure (contrast), noisiness (flatness and HNR), upper spectral extent (rolloff), temporal change (flux), and harmonic regularity (inharmonicity). The mean/standard-deviation pairs are intentional only where time variation adds a distinct interpretation. This set is explainable and broad, but should be described as a compact operationalization of timbre—not a complete representation of every perceptual timbre dimension.

4. Harmony branch — stable 12

The harmony branch starts from pitch-class/chroma representations, which intentionally fold octave-equivalent pitches into 12 pitch classes and are widely used for chord recognition, tonal analysis, alignment, structure discovery, and retrieval [6]. Tonal-centroid (Tonnetz) geometry then converts pitch-class relations into distances and trajectories that expose harmonic motion [7].

Feature	What it measures	Readable meaning	Evidence and study fit
global_key_strength	Strength of the best global key-profile match.	How clearly the track supports one global key.	Key-profile matching is a foundational computational model of key finding [8,9].
mode_confidence	Confidence/separation for major versus minor interpretation.	How unambiguous the inferred mode is.	Key-finding research treats tonic and mode as profile-matching decisions; confidence makes ambiguity explicit [8,9].
key_stability	Consistency of local key evidence through time.	Stable tonality versus modulation or uncertain local key.	Temporal key evidence and changes are central limitations/extensions of global key models [9].
chroma_entropy_mean	Mean Shannon entropy of framewise pitch-class energy.	Concentrated pitch-class content versus diffuse/complex tonal content.	Chroma entropy is used to quantify tonal complexity from local pitch-class distributions [10].
chroma_entropy_std	Variability of chroma entropy over time.	Whether tonal complexity is uniform or alternates across sections.	A transparent temporal statistic over the established entropy-based complexity measure [10].
chroma_flux_mean	Mean frame-to-frame change in chroma.	Average harmonic/pitch-class activity.	Chroma is an established harmonic representation [6]; temporal change of chroma operationalizes harmonic activity and complements geometric Tonnetz motion.
tonnetz_movement_mean	Mean distance between successive tonal-centroid positions.	Average harmonic movement between nearby frames.	Harte et al. introduced tonal-centroid movement for detecting harmonic change [7].
tonal_centroid_radius_mean	Mean distance from the tonal-centroid origin.	Degree of tonal concentration in Tonnetz space.	Tonal-centroid geometry represents harmonic relations; radius is an interpretable geometric summary of concentration [7].
chroma_recurrence_15s	Similarity/recurrence of chroma patterns at a 15-second lag.	Return of harmonic material at a musically meaningful section-scale lag.	Chroma is widely used for music structure and recurrence analysis [6]; this statistic summarizes repeated harmonic content.
chord_strength_mean	Mean confidence/strength of chord-template matches.	Clarity of detected chordal harmony.	Essentia exposes chord strength and chord descriptors within its tonal analysis framework [11,12].
chord_change_rate	Rate of changes in the estimated chord sequence.	Harmonic rhythm: how frequently chords change.	Chord change rate is an established Essentia high-level tonal descriptor [11,12].
chord_vocabulary_ratio	Unique detected chord classes relative to chord observations.	Diversity of the track's chord vocabulary.	Chord distributions/progressions carry genre information; language-model studies demonstrate genre discrimination from chord structures [13].

What the stable harmony vector can and cannot say

The chord-aware features improve sensitivity to progression-related behavior: chord-change rate captures harmonic rhythm, vocabulary ratio captures chord diversity, and chord strength captures reliability/clarity. However, the 12-dimensional vector does not encode the exact ordered chord progression (for example, I–V–vi–IV). It summarizes progression behavior. Exact sequence modeling would require a chord-sequence model and is deliberately outside this compact explainable branch.

5. Harmony warehouse — 15 retained fields

All 15 fields remain available for audit, ablation, correlation analysis, and future model versions. Only 12 are authorized for the current stable harmony branch.

Warehouse feature	Current status	Interpretive role
global_key_strength	Stable 12	Global key clarity
mode_confidence	Stable 12	Major/minor certainty
key_stability	Stable 12	Temporal key consistency
chroma_entropy_mean	Stable 12	Average tonal complexity
chroma_entropy_std	Stable 12	Variability of tonal complexity
chroma_flux_mean	Stable 12	Average pitch-class change
chroma_flux_std	Warehouse only	Variability of pitch-class change
tonnetz_movement_mean	Stable 12	Average harmonic movement
tonnetz_movement_std	Warehouse only	Variability of harmonic movement
tonal_centroid_radius_mean	Stable 12	Average tonal concentration
tonal_centroid_radius_std	Warehouse only	Variability of tonal concentration
chroma_recurrence_15s	Stable 12	Section-scale harmonic recurrence
chord_strength_mean	Stable 12	Chord clarity
chord_change_rate	Stable 12	Harmonic rhythm
chord_vocabulary_ratio	Stable 12	Chord diversity

Warehouse-only fields excluded from the stable vector: chroma_flux_std, tonnetz_movement_std, and tonal_centroid_radius_std. Their exclusion prevents three additional variability statistics from silently changing the registered 12-dimensional model interface; they remain available for later ablation evidence.

6. Rhythm branch — AcousticBrainz/Essentia provenance

The Rhythm-10 vocabulary is based on the AcousticBrainz feature schema and its Essentia extractor. AcousticBrainz was introduced as a community platform for collecting reproducible audio descriptors [14], while Essentia's Music Extractor explicitly provides onset, beat, BPM, BPM-histogram, beat-loudness, and danceability descriptors [11,12]. This provenance improves methodological credibility because the branch uses established MIR concepts rather than ad hoc labels.

Stable feature	Interpretation
bpm	Estimated global tempo
beats_count	Number of detected beat events
beats_loudness_mean	Mean energy/loudness around detected beats
bpm_histogram_first_peak_bpm	Tempo represented by the dominant BPM-histogram peak
bpm_histogram_first_peak_spread	Uncertainty/spread around the dominant tempo peak
bpm_histogram_first_peak_weight	Relative prominence of the dominant tempo peak
onset_rate	Rate of detected note/event onsets
danceability	Essentia rhythm-based danceability estimate
beat_interval_mean	Mean inter-beat interval derived transparently from beat positions
beat_interval_std	Variability of inter-beat intervals derived from beat positions

Precision note. The first eight names align directly with AcousticBrainz/Essentia rhythm outputs or their documented statistics. beat_interval_mean and beat_interval_std are transparent project-level summaries derived from the extractor's beat positions; they should be described as AcousticBrainz/Essentia-informed derivatives, not as literal AcousticBrainz field names.

7. Instrument and genre branches

Instrument-41 represents interpretable source-presence concepts: accordion, acoustic bass guitar, acoustic guitar, bass, beat, bell, bongo, brass, cello, clarinet, classical guitar, computer, double bass, drum machine, drums, electric guitar, electric piano, flute, guitar, harmonica, harp, horn, keyboard, oboe, orchestra, organ, pad, percussion, piano, pipe organ, Rhodes, sampler, saxophone, strings, synthesizer, trombone, trumpet, ukulele, viola, violin, and voice.

Genre-6 contains the six multilabel targets. Genre labels are outputs, not acoustic explanation inputs. Keeping the roles separate prevents target leakage and makes branch-level explanations interpretable: instrument, rhythm, timbre, and harmony describe the audio; genre is what the model predicts.

8. Reproducibility and reporting rules

- Report the stable dimensionality exactly as 41 instrument + 10 rhythm + 12 timbre + 12 harmony, with 6 genre outputs.
- State that 15 harmony descriptors exist in the warehouse, but only the named 12 enter the stable harmony branch.
- Treat means as track-level central tendency and standard deviations as within-track temporal variability.
- Fit normalization parameters on the training split only; apply the learned transform unchanged to validation and test tracks.
- Describe the cited literature as construct and methodological support, not proof that no alternative feature set could perform better.
- Preserve feature names, extraction configuration, track IDs, split assignments, and code version with each produced dataset.

9. Paper-ready methods wording

We used a fixed explainable concept interface comprising 41 instrument, 10 rhythm, 12 timbre, and 12 harmony features, with six multilabel genre targets. The timbre descriptors span spectral centre and spread, spectral shape and contrast, noise/harmonic balance, temporal spectral change, and inharmonicity, drawing on established timbre and genre-classification literature [1–5]. The harmony descriptors combine key/mode evidence, chroma complexity and movement, Tonnetz geometry, recurrence, and chord-level summaries [6–13]. Rhythm descriptors follow the AcousticBrainz/Essentia feature vocabulary; the two inter-beat-interval statistics are transparent derivatives of detected beat positions [11,12,14]. Although 15 harmony descriptors are retained in the feature warehouse, only the prespecified Harmony-12 enter the stable model.

References

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Appendix A. Exact stable feature lists

Timbre-12: spectral_centroid_mean, spectral_centroid_std, spectral_bandwidth_mean, spectral_bandwidth_std, spectral_skewness_mean, spectral_contrast_mean, spectral_flatness_mean, spectral_rolloff_mean, spectral_flux_mean, spectral_flux_std, hnr_mean_db, inharmonicity_mean.

Harmony-12: global_key_strength, mode_confidence, key_stability, chroma_entropy_mean, chroma_entropy_std, chroma_flux_mean, tonnetz_movement_mean, tonal_centroid_radius_mean, chroma_recurrence_15s, chord_strength_mean, chord_change_rate, chord_vocabulary_ratio.

Rhythm-10: bpm, beats_count, beats_loudness_mean, bpm_histogram_first_peak_bpm, bpm_histogram_first_peak_spread, bpm_histogram_first_peak_weight, onset_rate, danceability, beat_interval_mean, beat_interval_std.